

Curriculum vitae

Akke Mats Houben

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Personal data

NAME: Akke Mats Houben
E-MAIL: akke@akkehouben.net
akkemats.houben@ub.edu
ORCID: 0000-0002-8215-7882
DATE OF BIRTH: 10-04-1991
WEBPAGE: <https://www.akkehouben.net>

Current position

2021-present **Doctoral student in physics**
Departament de Física de la Matèria Condensada & Institute of Complex Systems
University of Barcelona, Barcelona, Spain
ADVISORS: Jordi Soriano & Jordi Garcia-Ojalvo
FUNDING: NEU-CHiP (Horizon 2020, 964877 – NEUCHIP)

Publications

1. **A.M. Houben**, A.C. Haeb, J. Garcia-Ojalvo & J. Soriano. Reservoir computing in simulated neuronal cultures: Effect of network structure. *Chaos: An Interdisciplinary Journal of Nonlinear Science* (2026)
2. **A.M. Houben**, J. Garcia-Ojalvo & J. Soriano. Role of connectivity anisotropies in the dynamics of cultured neuronal networks. *PLOS Computational Biology* (2025)
3. T. Sumi, **A.M. Houben**, ..., & A. Hirano-Iwata. Modular architecture confers robustness to damage and facilitates recovery in spiking neural networks modeling in vitro neurons. *Frontiers in Neuroscience* (2025)
4. H.F. Po, **A.M. Houben**, ..., D. Saad. Inferring structure of cortical neuronal networks from activity data: A statistical physics approach. *PNAS Nexus* (2025)
5. G. Menesse, **A.M. Houben**, J. Soriano & J.J. Torres. Integrated Information Decomposition Unveils Major Structural Traits of In Silico and In Vitro Neuronal Networks. *Chaos: An Interdisciplinary Journal of Nonlinear Science* (2024)
6. M. Montalà-Flaquer, ..., **A.M. Houben**, ... & J. Soriano. Rich dynamics and functional organization on topographically designed neuronal networks in vitro. *iScience* (2022)
7. **A.M. Houben**. Frequency selectivity of neural circuits with heterogeneous discrete transmission delays. *Neural Computation* (2021)
8. **A.M. Houben** & M.S. Keil. A calcium-influx-dependent plasticity model exhibiting multiple STDP curves. *Journal of Computational Neuroscience* (2020)

Book chapters

1. C.F. López-Leon*, **A.M. Houben***, ... & J. Soriano. Emergent complex dynamics in neuronal cultures and its relation to neuroengineering and medicine. In: *Nonlinear Dynamics for Biological Systems* (2025)

Preprints & submitted

1. H.F. Po, **A.M. Houben**, ... & D. Saad. Inferring Structure of Cortical Neuronal Networks from Firing Data: A Statistical Physics Approach. [arXiv:2402.18788](#)

Peer-reviewed proceedings

1. **A.M. Houben**, J. Garcia-Ojalvo & J. Soriano. Liquid State Computing in Neuronal Cultures: Effects of Connectivity Modularity on Response Separation and Generalisation in Numerical Simulations and Experiments. *IEICE Proceedings Series 76* (D1L-11) (2023)
2. H.F. Po, **A.M. Houben**, J. Soriano & D. Saad. Effective Structure Inference from Multi-Phase Cortical Neural Activities. *IEICE Proceedings Series 76* (D2L-15) (2023)
3. A.C. Haeb, **A.M. Houben**, J. Soriano & D. Tornero. Experiments on Modular Neuronal Cultures Monitored by High-Density Multielectrode Arrays. *IEICE Proceedings Series 76* (D1L-12) (2023)

Conference, seminar & workshop presentations

Talks

1. **A.M. Houben** & J. Soriano. Neural fields with auto-associative memories: collective activity, pattern formation and memory dynamics. *Dynamics Days Europe*, 23-27 June 2025
2. **A.M. Houben** & J. Soriano. Neural fields with auto-associative memories: collective activity, pattern formation and memory dynamics. *International conference on mathematical neuroscience (ICMNS)*, 17-20 June 2025
3. **A.M. Houben** & J. Soriano. Incompleteness of recurrent network dynamics: theoretical limitations to computation on neuronal networks. *BARCCSYN*, 30-31 May 2024
4. **A.M. Houben**, A.-C. Haeb, J. Garcia-Ojalvo & J. Soriano. Liquid state computing in neuronal cultures: effects of connectivity modularity on response separation and generalisation in numerical simulations and experiments. *2nd meeting of the Spanish chapter of the Complex Systems Society*, 22-23 February 2024
5. **A.M. Houben**, A.-C. Haeb, J. Garcia-Ojalvo & J. Soriano. Liquid state computing in neuronal cultures: effects of noise and connectivity modularity on response separation and generalisation in numerical simulations. *NOLTA*, 26-29 September 2023
6. **A.M. Houben**, A.-C. Haeb, J. Garcia-Ojalvo & J. Soriano. Liquid state computing in neuronal cultures: effects of noise and connectivity modularity on response separation and generalisation in numerical simulations. *BARCCSYN*, 25-26 May 2023
7. **A.M. Houben**, J. Garcia-Ojalvo & J. Soriano. Interplay of external noise and anisotropic connectivity on the dynamics of neuronal networks. *Mediterranean School of Complex Networks*, 25 June-02 July 2022

Short talks

1. **A.M. Houben** & J. Soriano. Incompleteness of recurrent network dynamics: theoretical limitations to computation on neuronal networks. *Maturation and plasticity in biological and artificial networks (IESC workshop)*, 21-25 October 2024

Seminar talks

1. **A.M. Houben**. Spatio-temporal pattern formation and dynamics in neuronal cultures. Consolider Seminar, University of Barcelona. 6 March 2025.

Poster presentations

1. **A.M. Houben** & J. Soriano. Neural fields with auto-associative memories: collective activity, pattern formation and memory dynamics. *BARCCSYN*, 22-23 May 2025
2. **A.M. Houben** & J. Soriano. Incompleteness of recurrent network dynamics: theoretical limitations to computation on neuronal networks. *Maturation and plasticity in biological and artificial networks (IESC workshop)*, 21-25 October 2024
3. **A.M. Houben**, A.-C. Haeb, J. Garcia-Ojalvo & J. Soriano. Computation with spontaneous input-responses in neuronal cultures. *FENS*, 25-29 June 2024
4. **A.M. Houben**, A.-C. Haeb, J. Garcia-Ojalvo & J. Soriano. Computation with spontaneous input-responses in neuronal cultures. *Cortical neurons for biological computing (FENS satellite)*, 24 June 2024
5. **A.M. Houben**, J. Garcia-Ojalvo & J. Soriano. Excitation-to-inhibition ratio determines pattern forming instability type in pulse-coupled phase oscillator populations. *International conference on mathematical neuroscience (ICMNS)*, 11-14 June 2024
6. **A.M. Houben**, A.-C. Haeb, J. Garcia-Ojalvo & J. Soriano. Computing in neuronal cultures: effects of noise and connectivity modularity on response separation and generalisation. *Symposium of the Barcelona Collaboratorium* 5-6 October 2023
7. **A.M. Houben** & M.S. Keil. Calcium influx dependent plasticity model. *BARCCSYN* 24-25 May 2018

Ad-hoc reviewer

Journals

- Journal of Computational Neuroscience

Conferences

- CompleNet, Zaragoza 2026
- NOLTA, Catania 2023

Teaching & advisory experience

Courses

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| 2014-2015 | Second-year C++ programming. Utrecht University of the Arts |
| 2014-2015 | First-year LISP & java programming. Utrecht University of the Arts |

Student advising

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| 2023 | Bachelor internship: Programming MEA data acquisition software |
| 2023 | Bachelor internship: Robot - neuronal culture interfacing |
| 2023 | Erasmus+ master thesis: Plasticity in neuronal culture simulations |
| 2022 | Bachelor internship: Numerical simulations of neuronal cultures |
| 2022 | Erasmus+ internship: Numerical simulations of neuronal cultures |

Education

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| 2017-2018 | Mres in behaviour and cognition. University of Barcelona, Barcelona, Spain |
| 2013-2017 | Ba (Hons.) in music and technology. Utrecht University of the Arts, Utrecht, The Netherlands |

Additional courses

- 2017 **Neuronal noise and neural signals.** Berstein Center for Computational Neuroscience, Berlin, Germany
- 2017 **Machine intelligence II.** Technical University of Berlin, Berlin, Germany
- 2016 **Machine learning.** Utrecht University, Utrecht, The Netherlands
- 2016 **Logic.** Utrecht University, Utrecht, The Netherlands

Working experience

- 2021-present **Doctoral student in physics.** University of Barcelona, Barcelona, Spain
- 2024-2025 **Biomedical data analyst.** University of Barcelona, Barcelona, Spain
- 2019-2021 **Software engineer.** nTh Netrowk Technologies to Help, Padova, Italy
- 2019-2020 **Neuronal data analyst.** Università degli Studi di Milano-Bicocca, Milan, Italy
- 2016-2017 **Software developer.** DDL Diagnostics Laboratory B.V., Rijswijk, The Netherlands
- 2015-2016 **Research intern.** Universitat Pompeu Fabra, Barcelona, Spain
- 2014-2015 **Student delegate at participation council.** Utrecht University of the Arts, Utrecht, The Netherlands

Outreach & media

1. Magazine article: Exploring the Intersection of Neuronal Cultures, Robotics and AI. *EE Times*, 23 April 2024 (<https://www.eetimes.com/exploring-the-intersection-of-neuronal-cultures-robotics-and-ai/>)

Student awards & grants

- 2017 Double honors bachelor studies. Utrecht University of the Arts, Utrecht, The Netherlands
- 2017 Jan van Scoral exchange grant. Utrecht University of the Arts, Utrecht, The Netherlands
- 2015-2016 Erasmus+ traineeship grant. Utrecht University of the Arts, Utrecht, The Netherlands

Other skills & competencies

- Languages:** Dutch (mother tongue)
 English (C2)
 Italian (B1)
 German (B1)
 Spanish (A2)
 Catalan (A1)